

Data Structures and Algorithms in Your Program

What the official regulations say — extracted, with page numbers

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This document extracts **everything the official regulations say about Data Structures and Algorithms**, so you do not have to read 328 pages of Arabic bylaw to find out what your course is.

Every fact below carries its **file and page number**. Go and check them.

Which document applies to you?

Document	Year	Covers	Status
FCIS-Bylaw-2026.pdf	2026	All seven programs	Authoritative
Program-ArtificialIntelligence-2020.pdf	2020	AI only	Superseded
Program-MedicalInformatics-2014.pdf	2014	Medical Informatics only	Superseded
Program-SoftwareEngineering-2013.pdf	2013	Software Engineering only	Superseded

If you enrolled under the 2026 bylaw, read the first row and ignore the rest. The older bylaws are kept because students still completing under them need them, and because it is useful to see how the course changed.

A note on the name “Bioinformatics”. There is no Bioinformatics program at the faculty. The program often called that is **Medical Informatics** — **المعلوماتية الطبية (MI)**, one of the seven programs in the 2026 bylaw. Bioinformatics is a course inside it: CS305 Bioinformatics in the 2026 bylaw (p. 39), MED121 in the 2014 one (p. 27). If you were told you are in “Bioinformatics”, the bylaw that governs you is the Medical Informatics one.

1. The course today — CS201

Source: FCIS Bylaw 2026, p. 138 (course specification) and p. 36 (study plan).

Field	Value
Code	CS201
English name	Data Structures and Algorithms
Arabic name	هياكل البيانات والخوارزميات
Credit hours	3
Lecture / Tutorial / Laboratory	2 / — / 2
Prerequisite	Introduction to Programming and Problem Solving (CS101)
Level / semester	الفصل الثاني (Sophomore) — الفصل الدراسي الأول (Fall)
Required student workload (SWL)	155 hours
Equivalent ECTS	5.14
Major in	IS, CS, IT, SE, MI, AI, NCS
Minor in	—
Targeted SDGs	4, 8, 9, 10, 11, 16, 17

Assessment weights (p. 138)

SA / PE	Midterm	Oral	Final
20	15	5	60

Course content, quoted verbatim (p. 138)

“This course introduces fundamental data structures, algorithms, and abstract data types using a mix of programming and theory. Main topics include data structures such as arrays, lists, linked lists, stacks, queues, hash tables, heaps, priority queues, graphs, and trees. It introduces algorithms used for list manipulation, graph searches, sorting, searching, and tree traversals. Also, it introduces analyzing and managing the complexity associated with data structures and their operations.”

What CS201 unlocks

Course	Arabic	When	Source
IS202 Introduction to Database Systems	مقدمة لنظم قواعد البيانات	Sophomore Spring	p. 33
CS303 Analysis and Design of Algorithms	تحليل وتصميم الخوارزميات	Junior Fall	p. 39, p. 144
CS305 Bioinformatics	—	CS major	p. 39
CS323 Healthcare Informatics	معلوماتية الرعاية الصحية	elective	p. 41
CS324 Biomedical Informatics	المعلوماتية الطبية	elective	p. 41
CS410 Algorithmic Graph Theory	نظرية الرسوم البيانية الحسابية	elective	p. 41
CS421 Biometrics	المقاييس الحيوية	elective	p. 42

CS303 — the sequel (p. 144)

CS303 Analysis and Design of Algorithms, 3 CH, 2 lecture + 2 lab, prerequisite **CS201**, Junior Fall, major in CS and AI, minor in NCS, same assessment split (20/15/5/60). It covers divide-and-conquer, recursion, greedy, **dynamic programming**, recursion trees, the master method, graph theory, NP-completeness, sorting and searching lower bounds, minimum spanning trees, shortest paths and network flows.

This is why dynamic programming is enrichment in CS201, not examinable material: the bylaw puts it in CS303.

2. Faculty-wide rules that apply to CS201

Source: FCIS Bylaw 2026.

Rule	Value	Page
Credit hours to graduate	142	p. 15–16
Minimum passing grade in a course	D	p. 21
Minimum CGPA to graduate	2.00 / 4.00	p. 15
Course pass rule	≥ 60% overall AND ≥ 30% of the final exam	p. 23 (Art. 17)
Attendance required to sit the final	≥ 75% of lectures and tutorials	p. 21 (Art. 15)
Penalty below that	محروم — barred, recorded as fail	p. 21
Final written exam	60% of the mark, 2 hours	pp. 23–24
Coursework (أعمال فصلية)	40% = 15 midterm + 20 practical/periodic + 5 oral	pp. 23–24

Rule	Value	Page
Semester length	16–17 weeks (summer 7–8)	pp. 14–15
Registration load	12–18 CH; 21 if CGPA ≥ 3 ; 12 if CGPA < 1	p. 18
Language of instruction	English	p. 14

Level thresholds (p. 17)

Level	Name	Credit hours completed
الأول	Freshman	0 – < 30
الثاني	Sophomore ← CS201	30 – < 66
الثالث	Junior	66 – < 99
الرابع	Senior	99 – < 142

Grade scale (p. 22)

A+ 4.0 ($\geq 97\%$) · A 4.0 (93–<97) · A– 3.7 (89–<93) · B+ 3.3 (84–<89) · B 3.0 (80–<84) · B– 2.7 (76–<80) · C+ 2.3 (73–<76) · C 2.0 (70–<73) · C– 1.7 (67–<70) · D+ 1.3 (64–<67) · D 1.0 (60–<64) · **F 0.0 (<60%)**

3. The same course under the older bylaws

Useful if you are completing under a previous bylaw, and interesting either way.

	2026 faculty	2020 AI	2014 Med. Inf.	2013 SW Eng.
Code	CS201	CS2101	IS122	IS122
Arabic name	هياكل البيانات والخوارزميات	تراكيب البيانات والخوارزميات [†]	هياكل البيانات وتحليل الخوارزميات	هياكل البيانات وتحليل الخوارزميات
Credit hours	3	3	3	3
Lec / Tut / Lab	2 / — / 2	2 / — / 2	2 / — / 2	2 / — / 2
Prerequisite	CS101 Intro to Programming	CS1002 Object Oriented Programming	CS012, MATH012	CS012, MATH012
Level	Sophomore Fall	Level 2, Semester 3	Level 2 (inferred [‡])	Sophomore, Semester 1
CH to graduate	142	138	135	135
Assessment	20/15/5/60	final $\leq 60\%$	final $\leq 60\%$	final $\leq 60\%$
Spec page	p. 138	p. 44	p. 35	p. 38
Plan page	p. 36	p. 34	(none [‡])	p. 29

† The 2020 AI bylaw uses **two different Arabic names** for the same course: تراكييب البيانات and الخوارزميات in the requirements table (p. 27) and هياكل البيانات و الخوارزميات in the course specification (p. 44).

‡ The 2014 Medical Informatics bylaw contains **no semester-by-semester study plan**, so it never states which semester IS122 is taken in. Level 2 is inferred from the course-code numbering rule on its p. 20, and matches the Software Engineering bylaw's explicit placement.

The older course description (2013, 2014 and 2020 bylaws)

The 2013 and 2014 bylaws share one description word for word:

"Introduce the fundamental concepts of data structures and the algorithms that proceed from them. Topics include recursion, the underlying philosophy of object-oriented programming, fundamental data structures (including stacks, queues, linked lists, hash tables, trees, and graphs), the basics of algorithmic analysis, and an introduction to the principles of language translation."

The 2026 description drops "the principles of language translation" and adds **heaps and priority queues** explicitly. The course got slightly more focused.

4. Known contradictions in the source documents

These are real defects in the official PDFs. They are recorded here rather than silently resolved. **In every case, the 2026 faculty bylaw governs.**

1. **Value of A+.** The 2013 SW Eng. bylaw gives **4.33** on p. 14 but **4.00** on p. 57 — inside the same document. The 2014, 2020 and 2026 bylaws all give **4.00**. → Use **4.00**.
 2. **Final exam weight, "لا تقل" vs "لا تزيد"**. The 2013 bylaw p. 13 says the final exam mark shall be **not less than** 60%; the 2014 bylaw p. 12 says **not more than** 60%. Both documents' own articles on p. 11 say no single component may *exceed* 60%, so the 2013 wording appears to be a typo. → The 2026 bylaw fixes this: final = **exactly 60**, p. 138.
 3. **B– grade band.** Listed under جيد جداً in one table and جيد in another, within both the 2013 and 2014 bylaws. → The 2026 scale (p. 22) is the one to use.
 4. **Two Arabic names for CS2101** in the 2020 AI bylaw — see † above.
 5. **Degraded scan.** Pages 41–43, 45–46, 48–50, 53 and 55–58 of the 2020 AI bylaw are a poor OCR layer with mangled course codes (AI3001 printed as A13001, UN1001 as UNI001). Pages 44, 47, 51 and 52 are clean. If you need a number from that document, prefer the clean pages or the 2026 bylaw.
 6. **Program naming.** The file published as bio_.pdf is the **Medical Informatics** bylaw, not a Bioinformatics one. See the note at the top.
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5. Where these files came from

Downloaded 22 September 2026 from the faculty website and stored here **byte-for-byte unmodified** (SHA-256 verified after copying); only the filenames were made descriptive.

File here	Original URL
FCIS-Bylaw-2026.pdf	https://csifac.mans.edu.eg/images/PDF/new/list-2026.pdf
Program-ArtificialIntelligence-2020.pdf	https://csifac.mans.edu.eg/images/PDF/program-work-regulations/ai-.pdf
Program-MedicalInformatics-2014.pdf	https://csifac.mans.edu.eg/images/PDF/program-work-regulations/bio_.pdf
Program-SoftwareEngineering-2013.pdf	https://csifac.mans.edu.eg/images/PDF/program-work-regulations/sw.pdf